

Final Year Project Document

Multi-Agent Market Research Platform

Aroon Kumar (2312466)
Mudasir Ali (2312465)
Abdullah Zafar Khan (2312467)

Supervised by: Mr. Zubair Ahmed

Department of Computer Science
Faculty of Computing and Engineering Sciences
SZABIST University
SZABIST (Islamabad Campus), Pakistan

Fall 2026

Multi-Agent Market Research Platform

By

Aroon Kumar (2312466)

Mudasir Ali (2312465)

Abdullah Zafar Khan (2312467)

A PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE (Bachelor of Computer Science)

(Supervisor) Mr. Zubair Ahmed	(FYP Committee Head) Mr. Muhammad Nadeem Khokhar

(Program Manager BS(CS))
Mr. Rana Faisal Hayat

Department of Department of Computer Science
Faculty of Computing and Engineering Sciences
Shaheed Zulfikar Ali Bhutto Institute of Science and Technology
SZABIST (Islamabad Campus), Pakistan

Fall 2026

DECLARATION

We, the candidates of Bachelor of Science (Computer Science) at Shaheed Zulfikar Ali Bhutto Institute of Science and Technology, Islamabad do hereby certify that this report titled **Multi-Agent Market Research Platform**, submitted as partial fulfillment of Bachelor of Science (Computer Science) degree requirements, is our original work and we are its sole author. All the employed materials, references to the literature and the work of others have been referred to and duly cited. This report has not been presented for examination anywhere else.

Aroon Kumar
2312466

Mudasir Ali
2312465

Abdullah Zafar Khan
2312467

Project Overview

A single application bridges the gap between onsite data collection and backend analysis. Survey teams gather responses, images, and audio directly in the field. Spatial coordinates get logged at the exact moment a worker hits capture, locking every photo to a precise physical location. If the internet drops out, the device simply caches the payload locally. Background syncing happens automatically later. Once data reaches the server, AI agents start scanning. They flag incomplete entries and extract the core metrics needed for research. Supervisors do not wait for delayed batch reports anymore. A live dashboard lets administrators validate individual surveys the second they appear in the database.

Analysts use a web portal for strategy. Surveyors use an Android app to log data. A single platform connects field operations with backend analysis. A database links media files to surveys. Investigators use this connection as a history log. A local cache manages network drops. The software syncs data after internet returns. An AI module parses media files. The system extracts metrics to accelerate drafting. The program flags anomalous entries. Administrators control system access through secure logins. The dashboard displays real-time progress updates. Staff members validate the records before delivery. The platform generates final reports in standard formats. Cloud servers store all approved records permanently.

The proposed system manages users, questionnaires, campaigns, attendance, data collection, synchronization, verification, market intelligence, media processing, agent orchestration, field performance, and analytics. The system controls access through user permissions and registered devices. The Android application collects responses, photographs, audio, GPS coordinates, and timestamps. The application stores submissions offline and synchronizes records when connectivity returns. The AI agents support research planning, information extraction, verification, analysis, and reporting. The system routes uncertain evidence to human reviewers and displays campaign comparisons through dashboards. The system also supports supervised coding assistance and workflow evaluation. The platform serves market research agencies, brand teams, researchers, analysts, and field staff. It aims to reduce repeated handling of information and make campaign evidence easier to organize, inspect, and compare. Managers gain visibility into field activity, submission completeness, and research findings, while analysts retain access to supporting sources and documented calculations. Human review preserves oversight of disputed records. Its intended impact is more traceable research and better-informed decisions, with benefits assessed through measured workflow performance, numerical accuracy, and source support.

Dedication

Our families handle the daily background stress for us. This massive support allows our team to focus strictly on writing code. The belief from our parents pushes us forward during overwhelming workloads. We share this final milestone directly with them. Their patience makes this entire software system possible for us.

The SZABIST University professors shape our approach to software development. Our mentors answer our most confusing technical questions. The teachers evaluate our messy early drafts for major flaws. Our team applies those harsh critiques directly to the system architecture. We carry this practical guidance into our professional careers.

Close friends face these brutal university deadlines alongside our team. Our classmates stay in the computer labs until midnight with us. This constant companionship makes us feel much less isolated. Classmates trade ideas and fix software bugs with us daily. That shared frustration easily becomes our favorite university experience.

Field surveyors give this system an actual purpose for us. Data analysts provide our team with real-world inspiration. We write the backend synchronization algorithms, but the active field workers dictate the actual software requirements. Our group observes these data collection workflows daily. We sincerely want our platform to simplify their remote reporting routines.

Acknowledgment

We received critical guidance from Mr. Zubair Ahmed throughout the development cycle of the proposed system. We encountered technical dead ends frequently during the initial coding phases. His strict feedback consistently pulled our team out. He reviewed every single page of our messy documentation drafts. We applied his specific industry knowledge to overhaul the software architecture. We built a much better final system because of his dedication to our progress. Our team remains genuinely thankful for his valuable time.

Our team extended deep gratitude to SZABIST University. The Islamabad Campus provided critical infrastructure for us during development. The Computer Science testing labs hosted our group for several weeks. We utilized the available technical resources on site to compile heavy models. External evaluators offered harsh critiques to us after examining the broken early prototypes. We redesigned the system entirely from that feedback. We needed this specific campus access to finish the proposed system.

Several open-source repositories provided the necessary raw data for us. Independent sports academies supplied massive archives of cricket footage to our team. We fed those specific video files directly into the computer-vision models. The detection algorithms required thousands of testing frames from us to achieve baseline accuracy. These external communities saved our development timeline. We genuinely appreciated their extensive public datasets.

Close friends and family members kept us sane during this intense development cycle. Late-night software debugging caused severe burnout for our team. Relatives frequently stepped in to manage our outside responsibilities. This emotional backing gave us strength during the hardest weeks. Classmates prevented total exhaustion by forcing us away from the monitors. We ultimately delivered a functioning system because this support network refused to let us quit.

Revision History

Compiled By	Checked By	Date	Description	Version
Aroon Kumar	Mr. Zubair Ahmed	15 Sept 2026	Section 1	1.0
Mudasir Ali	Mr. Zubair Ahmed	15 Sept 2026	Section 2	1.1
Abdullah Zafar Khan	Mr. Zubair Ahmed	15 Sept 2026	Section 3	1.2
Aroon Kumar	Mr. Zubair Ahmed	16 Sept 2026	Section 4	1.3
Mudasir Ali	Mr. Zubair Ahmed	16 Sept 2026	Section 5	1.4
Abdullah Zafar Khan	Mr. Zubair Ahmed	16 Sept 2026	Section 6	1.5
Aroon Kumar	Mr. Zubair Ahmed	17 Sept 2026	Section 7	1.6
Mudasir Ali	Mr. Zubair Ahmed	17 Sept 2026	Section 8	1.7
Abdullah Zafar Khan	Mr. Zubair Ahmed	17 Sept 2026	Section 9	1.8
Aroon Kumar	Mr. Zubair Ahmed	18 Sept 2026	Section 10	1.9
Mudasir Ali	Mr. Zubair Ahmed	18 Sept 2026	Section 11	2.0
Abdullah Zafar Khan	Mr. Zubair Ahmed	18 Sept 2026	Section 12, 13	2.1

Contents

Dedication iii

List of Figures vii

List of Tables viii

List of Figures

List of Tables